

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

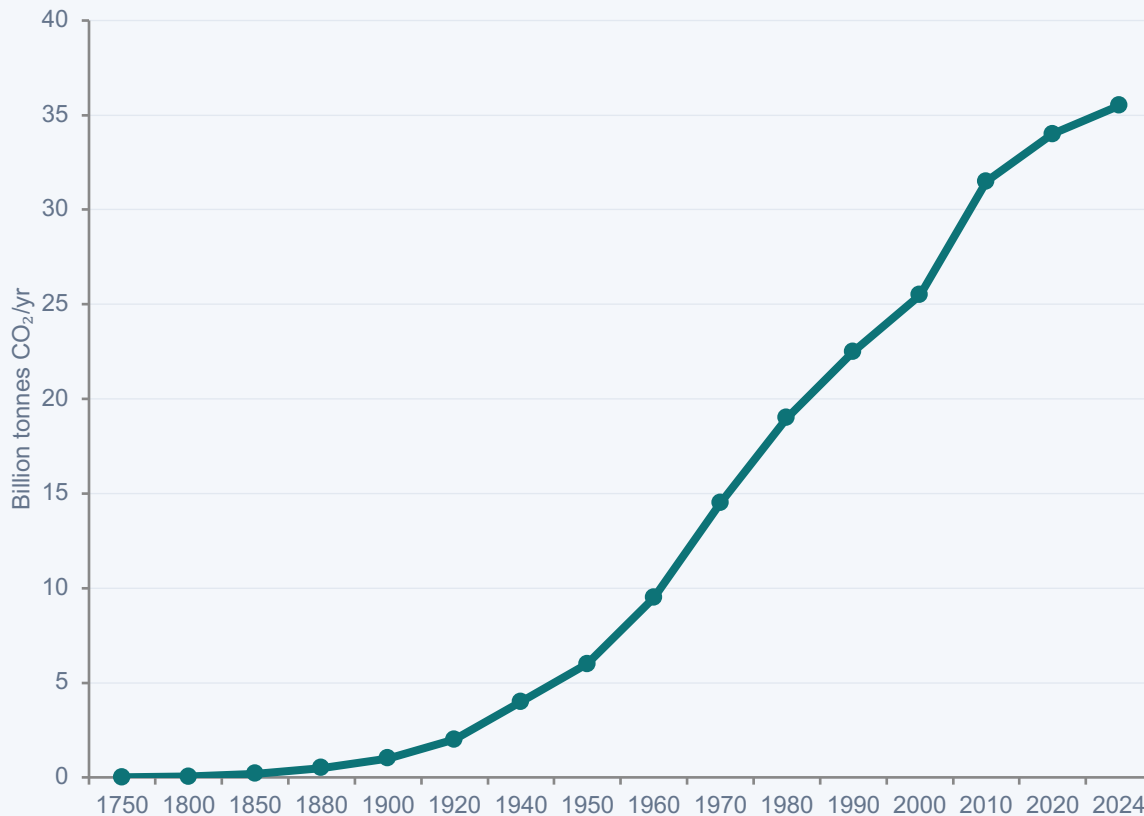
Fossil fuels & industry · 1750–2024

Data: Global Carbon Budget 2025 · Our World in Data · ourworldindata.org/co2-emissions

Prepared for policy analysts, climate negotiators, and sustainability researchers



Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



1950: ~6 Bt/yr

Post-WWII industrial growth begins

1990: ~22 Bt/yr

Nearly 4× 1950 levels

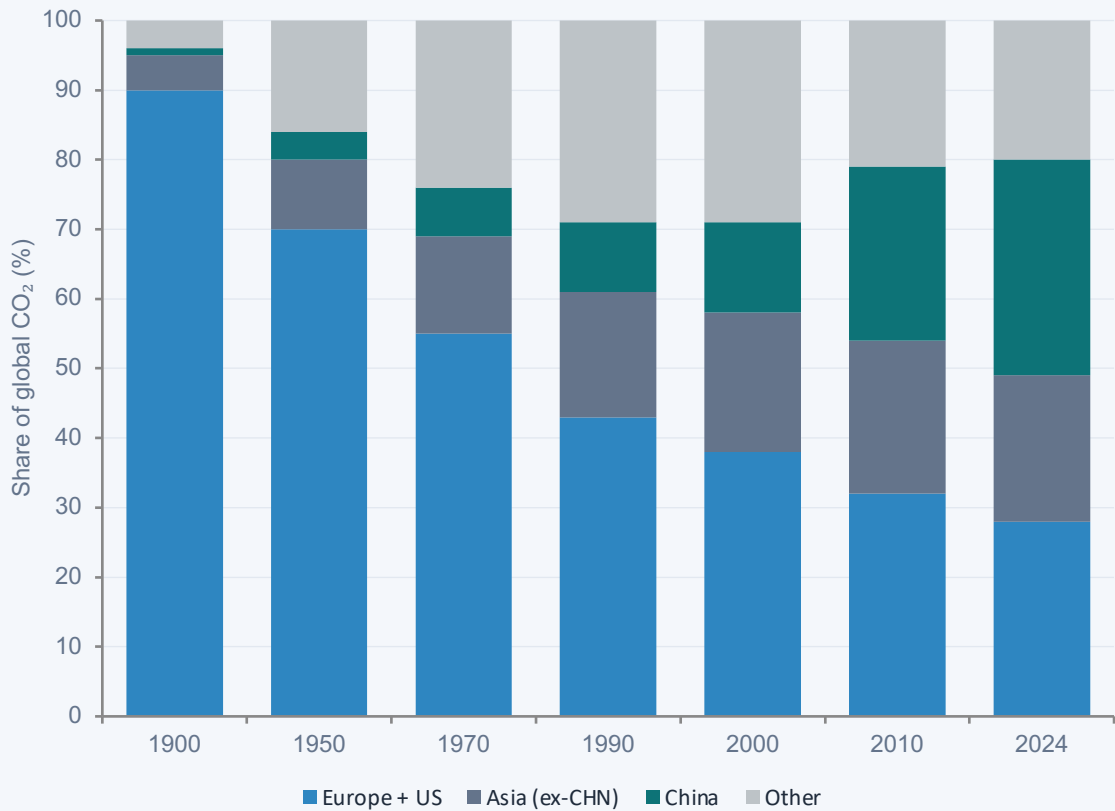
2024: >35 Bt/yr

Growth slowing — no peak yet

5.9×

increase since 1950

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



THE GREAT SHIFT

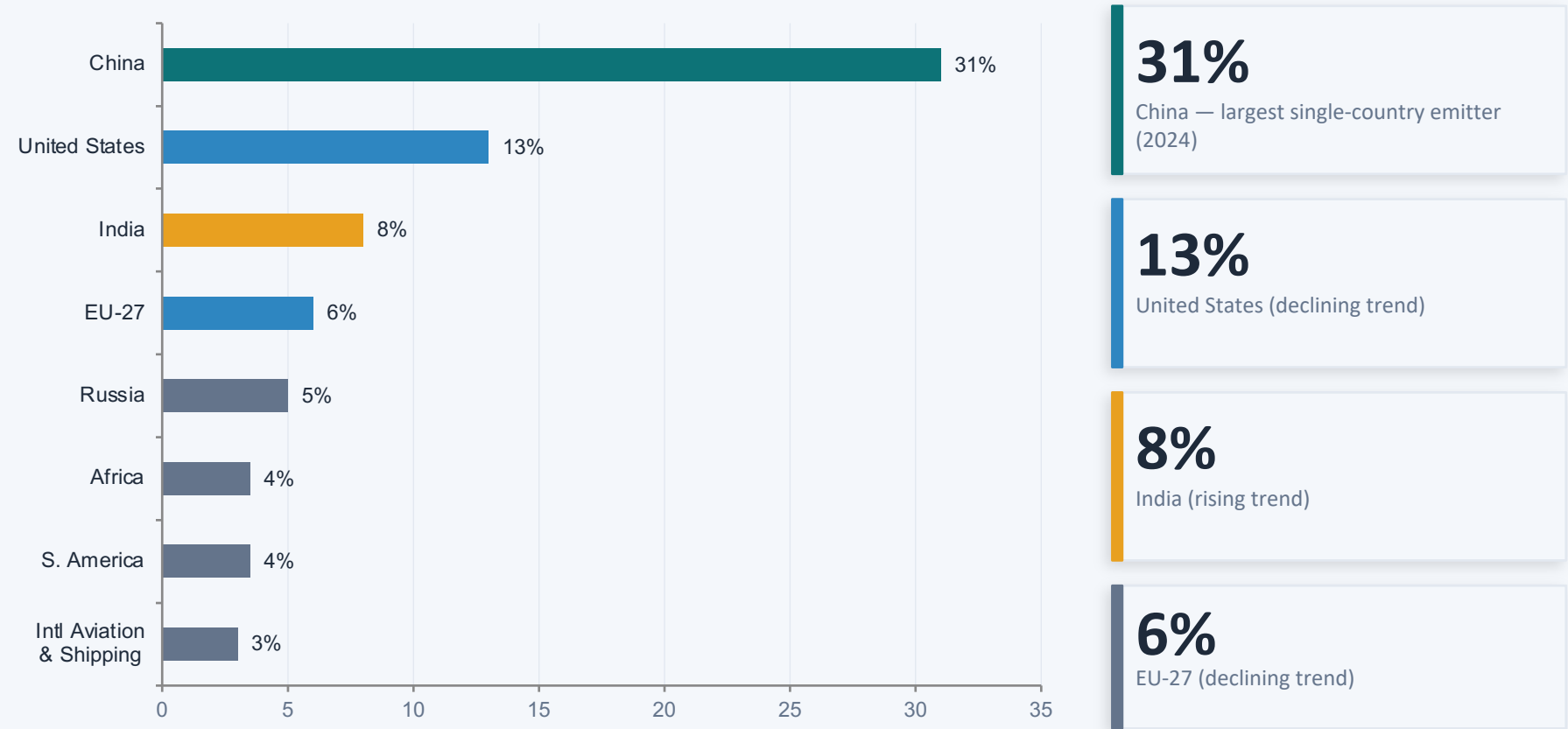
Year	EU+US	China	Asia*
1900	90%	<2%	~5%
1950	70%	4%	10%
2000	38%	13%	20%
2024	28%	31%	21%

*Asia excl. China

Asia now produces

~half of global emissions

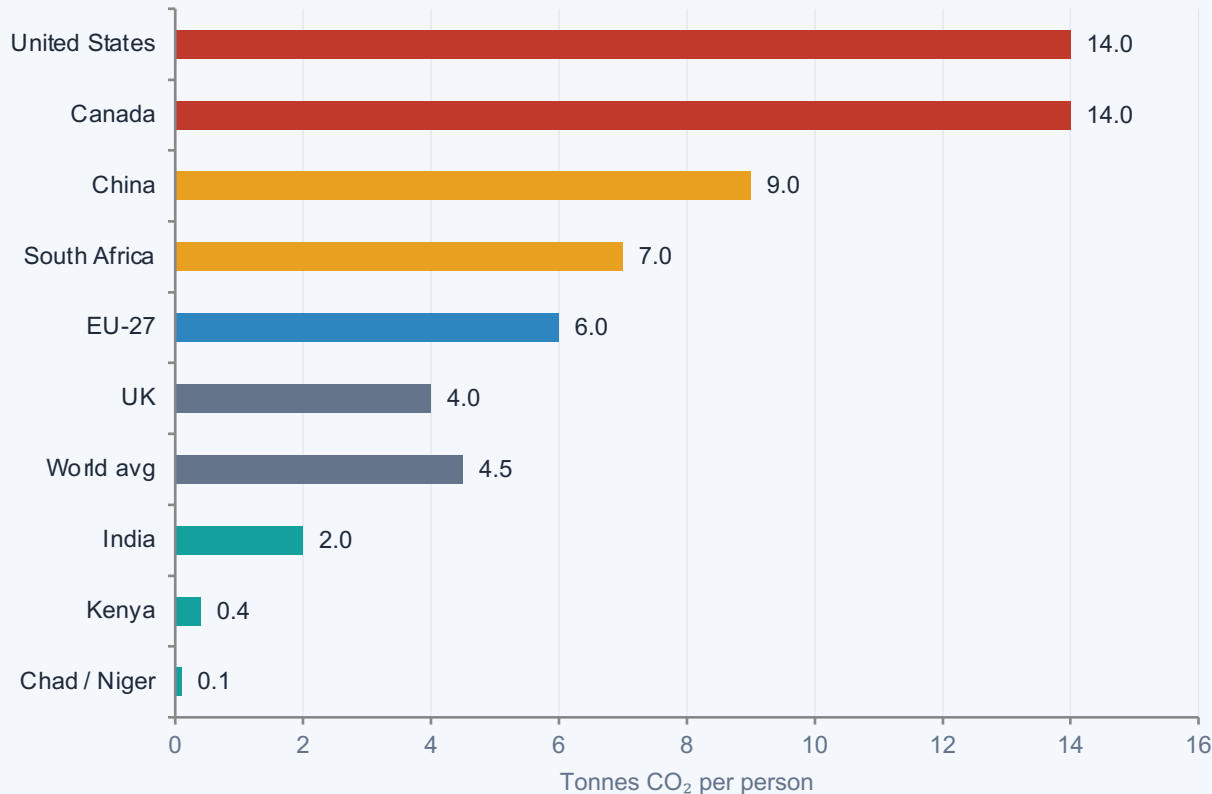
China Now Emits More Than One-Quarter of Global CO₂



Shares of global CO₂ emissions, 2024 · all countries/regions

Source: Global Carbon Budget 2025 via Our World in Data | ourworldindata.org/co2-emissions

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



150x

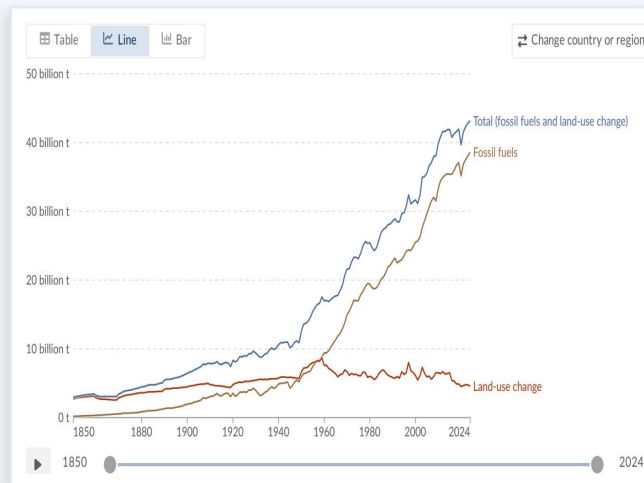
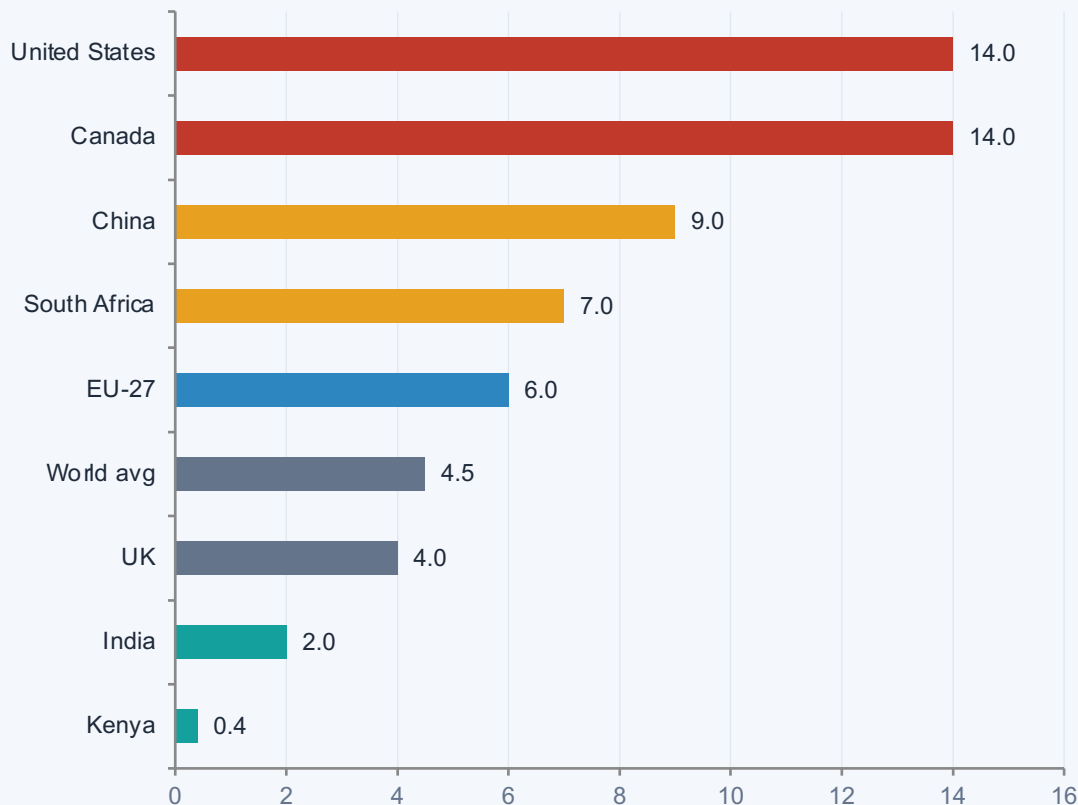
gap

between Chad (0.1 t) and US/Canada (~14 t)

Prosperity ≠ high emissions

France, UK & Portugal emit far less than Germany or the US despite similar living standards — thanks to nuclear and renewable energy.

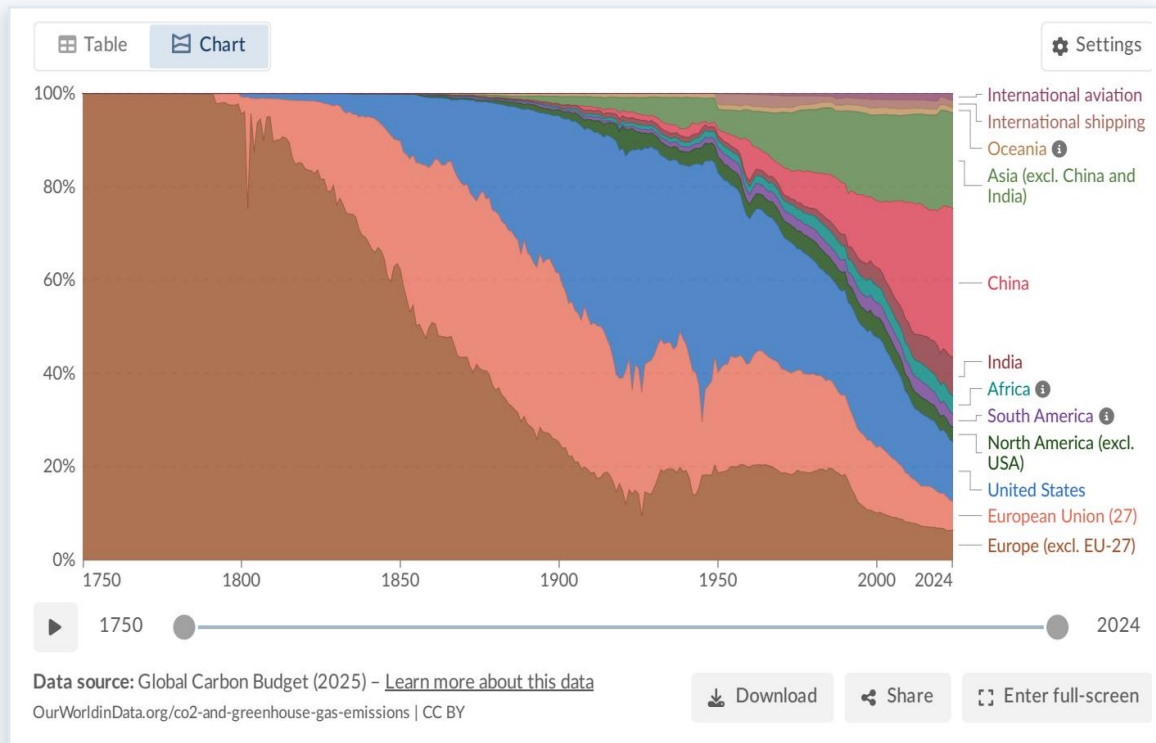
Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India



China is the #1 total emitter

yet its per capita (9 t) is well below the US (14 t) and Canada (14 t) — and only 2× the world average.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



Annual % change in CO₂ emissions, 2024 · Blue = decrease · Red/Orange = increase

2024 WINNERS & LAGGARDS

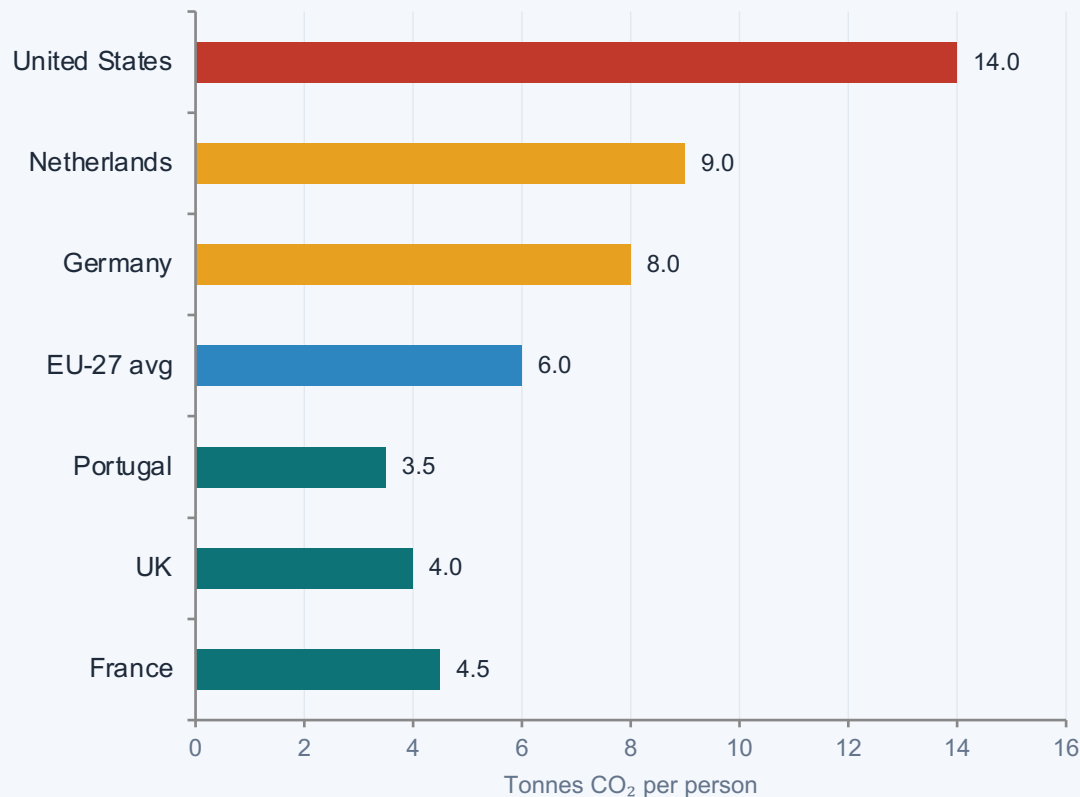
DECREASED

- ↓ United States
- ↓ European Union
- ↓ United Kingdom
- ↓ Germany
- ↓ Japan

INCREASED

- ↑ Russia
- ↑ SE Asia (parts)
- ↑ Central Asia (>10%)
- ↑ Parts of Africa

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



WHY THE DIFFERENCE?

Low-carbon leaders

France, UK, Portugal rely heavily on nuclear and/or renewables for electricity — keeping per capita emissions at 3.5–4.5 t.

EU average

The EU-27 as a whole (6 t) reflects a mix — with cleaner western members offset by coal-heavy eastern members.

Higher emitters

Germany, Netherlands, and the US still rely more on fossil fuels — resulting in 8–14 t per person.

Key lesson: similar prosperity levels can yield very different emissions depending on energy policy choices.

Key Takeaways

Responsibility Is Shared but Unequal

01

**Emissions still rising —
no peak yet**

Global CO₂ exceeds 35 Bt/yr. Growth has slowed but not reversed.

02

**China leads total; US
leads per capita among
large nations**

China emits 31% of global CO₂ but only ~9 t/person, well below the US (~14 t).

03

**Historical responsibility
skewed toward Europe
and the US**

They dominated emissions for over a century (>90% in 1900) before Asia's rise.

04

**Per capita inequality is
extreme: a 150× gap**

The average American emits in <2 days what a Chadian emits in a year.

05

**Energy policy can
decouple prosperity
from emissions**

France and the UK prove clean-energy economies are possible at high living standards.